

Emiliano M. Silva

Sustainable Mining Researcher

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Professional Summary

For the past 15+ years, I have worked in technical positions across chemistry, environmental geosciences, mining and construction engineering, and forensic science. I am currently a Ph.D. Candidate at the University of São Paulo, investigating methods to improve recovery in artisanal and small-scale mining operations. I hold B.Sc. degrees in Chemistry and Civil Engineering and a MSc in Geosciences, and I have worked as an Official Forensic Expert and Crime Scene Investigator for more than 12 years.

Although these roles may appear disconnected, I have applied the same rigorous approach across all of them: structured field procedures, reliable data collection, analytical reasoning, quality assurance, and clear technical reporting to solve complex practical problems. My experience also includes managing more than ten residential construction projects, strengthening my skills in planning, budgeting, resource management, and coordinating people, from construction workers to other Forensic Experts and Crime Scene Investigators, in multidisciplinary settings.

In my current research, I combine this practical background with international experience in Brazil, Colombia, Suriname, and Canada to develop sustainable approaches to mineral processing, mercury-free gold recovery, and mine-waste valorization. I am seeking to transition into a full-time international research career, where I can contribute this adaptable but methodologically consistent background to high-impact research in sustainable extractive metallurgy and related applied fields.

Research Interests

Sustainable Extractive Metallurgy • Hydrometallurgy and Mineral Processing • Mercury-Free Artisanal and Small-Scale Mining • Gold Leaching and Recovery • Environmental Geochemistry • Mine-Waste Valorization • Circular Economy • Critical Metals Recovery

Education

2023–Present **Ph.D. Candidate in Mineral Engineering**, *University of São Paulo (USP)*, São Paulo, Brazil

Research on mercury-free processing pathways for artisanal and small-scale mining, with emphasis on alternative leaching technologies, gold recovery, and sustainable mineral processing.

2020–2022 **B.Sc. in Civil Engineering**, *Faculdade Anhanguera*, Mato Grosso, Brazil

Final project investigated soil-cement bricks as a lower-impact material for sustainable construction.

2017–2018 **Specialization in Criminal Forensics and Forensic Sciences**, *Faculdade Poliensino*, Mato Grosso, Brazil

- 2010–2012 **M.Sc. in Geosciences**, *Federal University of Mato Grosso (UFMT)*, Mato Grosso, Brazil
Dissertation: *Statistical Analysis of Surface-Water Quality Parameters as Indicators of Selenium Geochemistry in a Canadian Coal Mine*. Research period at the University of British Columbia, Vancouver, Canada, supported by an ELAP scholarship.
- 2005–2009 **B.Sc. in Chemistry**, *Federal University of Mato Grosso (UFMT)*, Mato Grosso, Brazil
Undergraduate research focused on monitoring total mercury in fish from the Manso Hydroelectric Reservoir, Mato Grosso, Brazil.

Research Experience and International Collaboration

- 2023–Present **Researcher**, *NAP.Mineração, Research Center for Responsible Mining*, University of São Paulo, Brazil
Conduct research on sustainable artisanal and small-scale gold mining in Brazil, Colombia, and Suriname. Perform field diagnostics, environmental and process sampling, and data analysis to assess gold recovery, mercury losses, and resource efficiency. Contribute to mercury-free processing strategies and mine-waste valorization pathways.
- 2011 **Visiting Research Student**, *Norman B. Keevil Institute of Mining Engineering*, University of British Columbia, Vancouver, Canada
Conducted research in environmental geochemistry as an Emerging Leaders in the Americas Program scholarship recipient, supervised by Prof. Marcello Mariz da Veiga. Investigated water-quality parameters and selenium geochemistry in coal-mining environments.
- 2011 **Junior Geochemist, Trainee**, *Lorax Environmental Services Ltd.*, Vancouver, Canada
Contributed to the project *Role of Nitrate in the Remobilization and Attenuation of Selenium in Coal Mine Waste*, involving Lorax, the Mine Environment Neutral Drainage program, Natural Resources Canada, and industry partners. Analyzed surface-water and geochemical data from coal-mining operations.

Professional and Leadership Experience

- 2022–Present **Construction Engineer**, *Independent Practice*, Mato Grosso, Brazil
Manage the field execution, budgeting, resource allocation, procurement, and quality control of more than ten residential construction projects. Coordinate construction teams and suppliers, control costs and material use, and deliver practical solutions under operational constraints.
- 2014–Present **Official Forensic Expert and Crime Scene Investigator**, *POLITEC, State Secretariat of Public Security*, Mato Grosso, Brazil
Conduct official forensic examinations and crime-scene investigations that support criminal-justice proceedings and public-safety decision-making. Apply rigorous procedures for evidence collection, documentation, analysis, and chain-of-custody control, and prepare technically defensible reports for legal proceedings. Previously served as Regional Manager for the Juína and Primavera do Leste units, coordinating operations and liaising with judicial and regulatory authorities.

Teaching and Academic Service

- 2017–2020 **Part-time Lecturer and Program Coordinator**, *Faculdade Poliinsino*, Mato Grosso, Brazil
Coordinated and taught postgraduate specialization coursework in Criminalistics and Forensic Sciences, with emphasis on crime-scene processing and forensic investigative methodologies.
- 2012–2014 **Full-time Lecturer**, *Centro Universitário de Várzea Grande (UNIVAG)*, Mato Grosso, Brazil
Taught Statistics, Materials Science, General Chemistry, and Analytical Chemistry. Contributed to the institutional accreditation process for the Civil Engineering program.

- 2011–2014 **Guest Lecturer**, *State University of Mato Grosso (UNEMAT)*, Mato Grosso, Brazil
Delivered classroom and laboratory modules in General and Analytical Chemistry for chemistry degree students.
- 2010–2011 **Assistant Professor**, *Federal University of Mato Grosso (UFMT)*, Mato Grosso, Brazil
Taught Applied Analytical Chemistry and General Chemistry for Civil, Sanitary, and Environmental Engineering programs. Managed laboratory sessions and supported student training in chemical-analysis methods.

Core Competencies

	Level	Skill	Years	Comment
Technical	■■■■■	Field Operations	15+	<i>Forensic investigation, environmental and process sampling</i>
	■■■■■	Analytical Chemistry	15+	<i>Trace-element monitoring and environmental analysis</i>
	■■■■■	Data Analysis	12+	<i>Statistical analysis using R, Python, and Excel</i>
	■■■■■	Technical Reporting	12+	<i>Scientific and judicial documentation</i>
	■■■■■	ASGM Field Diagnostics	3+	<i>Gold-recovery assessment and mercury-loss evaluation</i>
Leadership	■■■■■	Forensic Unit Management	10+	<i>Regional operations, quality assurance, and institutional liaison</i>
	■■■■■	Construction Management	3+	<i>Budgeting, work oversight, suppliers, and construction teams</i>
	■■■■■	Teaching and Training	7+	<i>Chemistry, engineering, and forensic-science education</i>
Languages	■■■■■	Portuguese	Native	
	■■■■■	English	15+	<i>Duolingo English Test Score 130 (2026/07/03)</i>
	■■■■■	Spanish	10+	<i>Fieldwork communication</i>
	■■■■■	Italian	< 1	<i>Currently studying</i>

Selected Peer-Reviewed Publications

- Druiventak, A., Vriesde, E., Silva, E. M., & Veiga, M. M. (2026). From byproduct to business: A conceptual micro-enterprise model for artisanal and small-scale gold mining (asgm) tailings in suriname. *The Extractive Industries and Society*, 27, 101931. <https://doi.org/10.1016/j.exis.2026.101931>
- Silva, E. M., da Veiga, M. M., & De Tomi, G. (2026). System-wide evaluation of gold and silver recovery and mercury losses in an artisanal amalgamation–cyanidation circuit: A diagnostic approach to mercury-free processing. *Minerals*, 16(8), 784. <https://doi.org/10.3390/min16080784>
- Silva, E. M., Barreto, M. C. S., Veiga, M. M., & De Tomi, G. (2025). From agro-industrial waste to gold lixiviant: Evaluating cassava wastewater applications in artisanal mining. *Mining*, 5, 64. <https://doi.org/10.3390/mining5040064>
- Silva, E. M., Vriesde, E., Druiventak, A., Veiga, M. M., & De Tomi, G. (2025). Production of cement-tailings bricks with artisanal gold mining waste. *Cleaner Waste Systems*, 11, 100269. <https://doi.org/10.1016/j.clwas.2025.100269>

Veiga, M. M., Anene, N. C., & Silva, E. M. (2025). Four decades of efforts to reduce or eliminate mercury pollution in artisanal gold mining. *Minerals*, 15(4), 376. <https://doi.org/10.3390/min15040376>

Awards and Research Funding

- 2011 **Emerging Leaders in the Americas Program Scholarship**, *Canadian Bureau for International Education*, Canada
Supported research at the University of British Columbia, Vancouver, Canada.
- 2010 **M.Sc. Scholarship**, *National Council for Scientific and Technological Development (CNPq)*, Brazil
- 2009 **Full Funding Award for Attendance**, *9th International Conference on Mercury as a Global Pollutant*, Guiyang, China
- 2007–2009 **Undergraduate Research Scholarships**, *Institute of Mato Grosso Research (IPEM/UFMT)*, Brazil



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